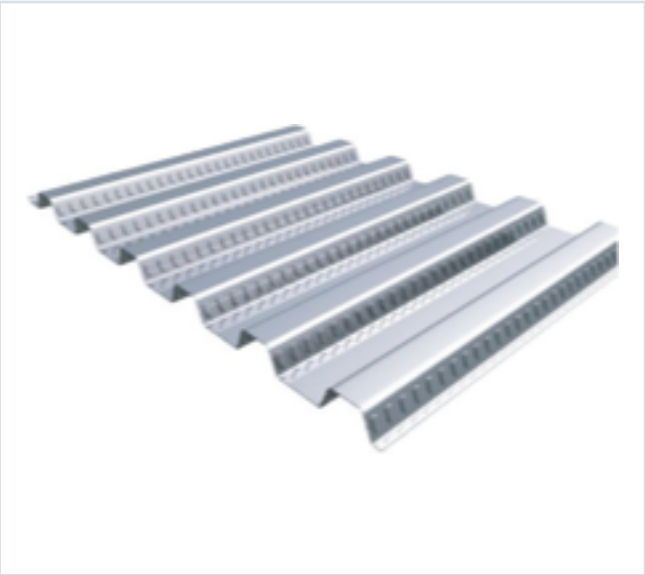


DECKING PRODUCT SUBMITTAL

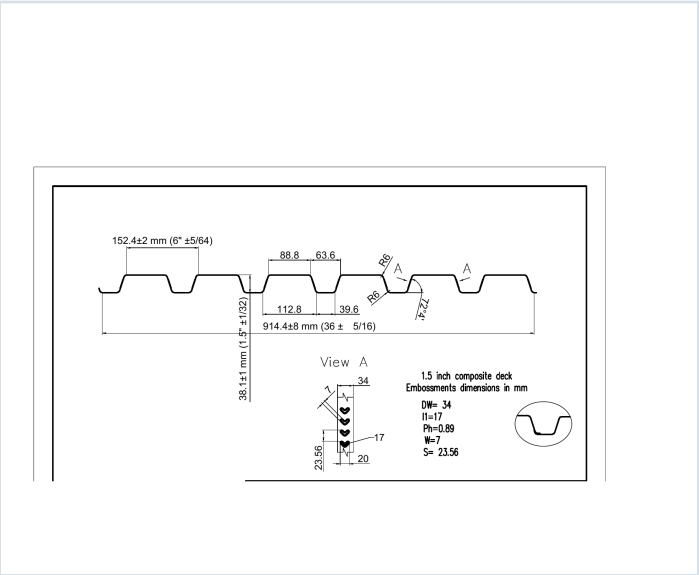
15CD-FY80-20-G60

PRODUCT	GRADE	GAUGE	COATING
1.5" Composite Deck	Grade 80	20 ga	G60

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 20 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0358	2.0	60	0.206	0.215	7398	7738	0.178	0.209

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 50, 51, 52, 53
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

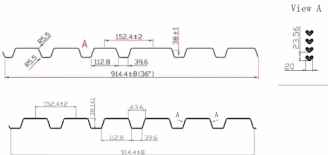
15CD-FY80-20-G60 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 50

GRADE 80 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (lbs)	E _x (in ⁴)	S _x (in ³)	S _y (in ³)	I _x (in ⁴)	I _y (in ⁴)	J _x (in ⁴)	J _y (in ⁴)
22	0.0395	16	60	0.166	0.175	7908	7738	0.162	0.167
20	0.0368	20	60	0.206	0.215	7908	7738	0.178	0.209
18	0.0341	24	60	0.246	0.255	7908	7738	0.212	0.248
16	0.0314	30	60	0.306	0.315	7908	7738	0.252	0.303

Note: All section properties and ASD flexural strengths are calculated in accordance with AISC 360-10, AISC 360-10.2, AISC 360-10.3, and AISC 360-10.4.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (R _n) (kips)	Web Crippling (R _n) (kips)
		One Flange Loading	Two Flange Loading
		End Bearing	Interior Bearing
22	2908	180	178
20	4563	172	172
18	6218	227	227
16	7873	307	307

Note: All section properties and ASD flexural strengths are calculated in accordance with AISC 360-10, AISC 360-10.2, AISC 360-10.3, and AISC 360-10.4.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	5'-0"	6'-0"	7'-0"	8'-0"	9'-0"	10'-0"
Single	22	197	163	137	117	101	88
	20	279	230	194	165	144	126
	18	359	297	249	212	186	164
	16	439	367	307	260	230	204
Double	22	167	138	116	99	86	74
	20	246	203	168	144	126	110
	18	325	268	221	190	168	146
	16	404	333	276	238	212	186
Triple	22	141	116	97	83	72	63
	20	210	175	146	125	109	95
	18	280	230	194	165	144	126
	16	349	289	243	208	183	161

Notes:
1. All section properties and ASD flexural strengths are calculated in accordance with AISC 360-10, AISC 360-10.2, AISC 360-10.3, and AISC 360-10.4.
2. Loads shown in tables are uniformly distributed superimposed loads in psf. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased for moment, clear span dimensions.
3. Working Moment Formula used for beam design: $M = \frac{wL^2}{8}$ for single span, $M = \frac{wL^2}{16}$ for two span, $M = \frac{wL^2}{24}$ for three span.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 51

Span	Gage	5'-0"	6'-0"	7'-0"	8'-0"	9'-0"	10'-0"
Single	22	197	163	137	117	101	88
	20	279	230	194	165	144	126
	18	359	297	249	212	186	164
	16	439	367	307	260	230	204
Double	22	167	138	116	99	86	74
	20	246	203	168	144	126	110
	18	325	268	221	190	168	146
	16	404	333	276	238	212	186
Triple	22	141	116	97	83	72	63
	20	210	175	146	125	109	95
	18	280	230	194	165	144	126
	16	349	289	243	208	183	161

Note: For loads that cause L/120 Deflection, multiply by 1.1. For loads that cause L/180 Deflection, multiply by 1.1. For loads that cause L/360 Deflection, multiply by 0.847.

Construction Span Table – 20 psf Construction Load

Total Slab Depth	Normal Weight Concrete (150 pcf)				Lightweight Concrete (115 pcf)			
	Deck Type	Maximum Unstressed Clear Span	Maximum Unstressed Clear Span	Maximum Unstressed Clear Span	Deck Type	Maximum Unstressed Clear Span	Maximum Unstressed Clear Span	Maximum Unstressed Clear Span
16 in. (1+200) 37 PSF	15x6x22 ga	8' 0"	8' 0"	8' 0"	15x6x22 ga	8' 0"	8' 0"	8' 0"
	15x6x18 ga	8' 0"	8' 0"	8' 0"	15x6x18 ga	8' 0"	8' 0"	8' 0"
	15x6x16 ga	8' 0"	8' 0"	8' 0"	15x6x16 ga	8' 0"	8' 0"	8' 0"
	15x6x14 ga	8' 0"	8' 0"	8' 0"	15x6x14 ga	8' 0"	8' 0"	8' 0"
400 (1+200) 43 PSF	15x6x22 ga	6' 0"	6' 0"	6' 0"	15x6x22 ga	6' 0"	6' 0"	6' 0"
	15x6x18 ga	6' 0"	6' 0"	6' 0"	15x6x18 ga	6' 0"	6' 0"	6' 0"
	15x6x16 ga	6' 0"	6' 0"	6' 0"	15x6x16 ga	6' 0"	6' 0"	6' 0"
	15x6x14 ga	6' 0"	6' 0"	6' 0"	15x6x14 ga	6' 0"	6' 0"	6' 0"
450 (1+200) 43 PSF	15x6x22 ga	6' 0"	6' 0"	6' 0"	15x6x22 ga	6' 0"	6' 0"	6' 0"
	15x6x18 ga	6' 0"	6' 0"	6' 0"	15x6x18 ga	6' 0"	6' 0"	6' 0"
	15x6x16 ga	6' 0"	6' 0"	6' 0"	15x6x16 ga	6' 0"	6' 0"	6' 0"
	15x6x14 ga	6' 0"	6' 0"	6' 0"	15x6x14 ga	6' 0"	6' 0"	6' 0"
500 (1+200) 43 PSF	15x6x22 ga	6' 0"	6' 0"	6' 0"	15x6x22 ga	6' 0"	6' 0"	6' 0"
	15x6x18 ga	6' 0"	6' 0"	6' 0"	15x6x18 ga	6' 0"	6' 0"	6' 0"
	15x6x16 ga	6' 0"	6' 0"	6' 0"	15x6x16 ga	6' 0"	6' 0"	6' 0"
	15x6x14 ga	6' 0"	6' 0"	6' 0"	15x6x14 ga	6' 0"	6' 0"	6' 0"
550 (1+200) 43 PSF	15x6x22 ga	6' 0"	6' 0"	6' 0"	15x6x22 ga	6' 0"	6' 0"	6' 0"
	15x6x18 ga	6' 0"	6' 0"	6' 0"	15x6x18 ga	6' 0"	6' 0"	6' 0"
	15x6x16 ga	6' 0"	6' 0"	6' 0"	15x6x16 ga	6' 0"	6' 0"	6' 0"
	15x6x14 ga	6' 0"	6' 0"	6' 0"	15x6x14 ga	6' 0"	6' 0"	6' 0"
600 (1+200) 46 PSF	15x6x22 ga	6' 0"	6' 0"	6' 0"	15x6x22 ga	6' 0"	6' 0"	6' 0"
	15x6x18 ga	6' 0"	6' 0"	6' 0"	15x6x18 ga	6' 0"	6' 0"	6' 0"
	15x6x16 ga	6' 0"	6' 0"	6' 0"	15x6x16 ga	6' 0"	6' 0"	6' 0"
	15x6x14 ga	6' 0"	6' 0"	6' 0"	15x6x14 ga	6' 0"	6' 0"	6' 0"

Note: Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

15CD-FY80-20-G60 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 52

22 ga Normalweight Concrete (145 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	37	400	400	375	374	209	232	202
4	37	400	400	400	397	340	293	238
4.5	43	400	400	400	400	400	397	311
5	49	400	400	400	400	400	400	369
5.5	55	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	177	166	136	123	110	98	89	80
4	224	187	175	166	159	125	113	102
4.5	273	241	213	190	170	153	138	125
5	323	289	255	228	202	182	164	148
5.5	375	331	294	262	235	211	191	175
6	400	378	336	300	269	242	218	197

Notes
1. AISI 901 C-2017 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.

20/18/16 ga Normalweight Concrete (145 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	37	400	400	400	382	327	283	246
4	37	400	400	400	400	400	364	312
4.5	43	400	400	400	400	400	400	381
5	49	400	400	400	400	400	400	400
5.5	55	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	236	181	170	151	136	122	110	100
4	276	242	215	192	172	155	142	127
4.5	333	296	263	236	211	190	172	156
5	390	352	313	280	251	226	205	186
5.5	450	400	364	325	292	263	238	216
6	400	400	400	372	334	301	272	247

Notes
1. Because of the profile of the embossments, there is no gain in strength for the composite deck slab when the deck gets thicker than 20 ga. However, the construction spans do get longer for 18 and 16 ga deck.
2. AISI 901 C-2017 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.

STEEL CATALOG PAGE 53

22 ga Lightweight Concrete (115 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	23	400	400	359	374	260	225	196
4	28	400	400	400	386	329	285	248
4.5	33	400	400	400	400	400	348	304
5	37	400	400	400	400	400	400	362
5.5	42	400	400	400	400	400	400	400
6	46	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	172	152	135	121	108	97	88	80
4	208	183	172	153	138	124	112	102
4.5	257	226	210	188	168	152	137	125
5	308	281	250	224	201	181	164	149
5.5	359	327	291	260	234	211	191	175
6	400	374	333	298	268	242	218	199

Notes
1. AISI 901 C-2017 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.

20/18/16 ga Lightweight Concrete (115 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	23	400	400	400	367	314	272	238
4	28	400	400	400	400	398	346	302
4.5	33	400	400	400	400	400	400	370
5	37	400	400	400	400	400	400	400
5.5	42	400	400	400	400	400	400	400
6	46	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	279	181	161	147	132	120	108	99
4	306	235	209	187	169	152	136	125
4.5	325	288	257	230	207	187	169	154
5	368	344	306	275	247	223	202	184
5.5	400	400	357	320	288	260	236	215
6	400	400	400	366	330	298	271	246

Notes
1. Because of the profile of the embossments, there is no gain in strength for the composite deck slab when the deck gets thicker than 20 ga. However, the construction spans do get longer for 18 and 16 ga deck.
2. AISI 901 C-2017 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.